

Course Code: ECE1002	Course Title: Fundamentals of Electrical and Electronics Engineering	TPC	3	2	4
Version No.	1.3				
Course Pre-requisites/ Co-requisites/ anti-requisites (if any).	NONE				
Objectives:	1. Analysis, presentation, and solution of DC and AC Circuits 2. To acquire knowledge about the characteristics and working of different semiconductor devices and their application 3. To acquire knowledge about the single-phase electrical machines, working principle, and construction				
CO's Mapping with PO's and PEO's					
Course Outcomes	Course Outcome Statement	PO's / PEO's			
CO1	Understanding the fundamental concept of DC Terminology, concept related to different transformation, Network Theorem.	PO1, PO2, PEO1			
CO2	Acquiring knowledge related to single phase system: concept of AC terminology, and AC power and extension to three phase system.	PO1, PO2, PO3, PEO1, PEO2.			
CO3	Understanding and analyzing Semiconductor Devices and Applications	PO2, PO3, PEO1, PEO2			
CO4	Understanding Operation Amplifier (Op-amps): Basics and applications	PO1, PO2, PO3, PO4, PO5, PEO1, PEO2			
CO5	Developing knowledge related to DC and AC Machine	PO1, PO2, PO4, PEO1			
		TOTAL HOURS OF INSTRUCTIONS: 45			
Module No. 1	DC Circuit Analysis	10 Hours			
DC Circuit Basics: Basic terminology recap (Conductors, Semiconductors, Insulators, Electric charge, Current, Resistance, Potential, Potential difference, Voltage, Work, Power, Energy – with units, Electric circuit, mesh, loop, node).Classification of circuit components (active, passive), Ohm’s law, Kirchoff’s law, nodal and mesh analysis, source transformation, star, and delta transformations. Cost of energy calculations. Sources (Voltage, Current – dependent and independent), R, L, C, and their calculations Circuit Theorems: Thevenin’s theorem, Superposition theorem. Maximum power transfer theorem.					
Module No. 2	AC Circuit Analysis	10 Hours			
Single-phase systems: Introduction, AC terminology, values (RMS Values, Average value, Form factor) of different signals. Concept of j-operator, impedance, and admittance. Power in					

AC circuits, problems related to calculation of active, reactive, and apparent power. Concept of power factor and its significance.

Three-phase systems: Introduction, Balanced systems, comparison between star and delta connection of loads. Calculation of three-phase voltage, current, power. Two wattmeter method for measuring three-phase power with examples.

Module No. 3	Semiconductor Devices and Applications	10 Hours
---------------------	---	-----------------

Conduction in Semiconductor materials; PN junction diodes: IV characteristics and current equation, Rectifier circuits design, and analysis; Zener diode and applications, BJT: BJT characteristics and biasing, BJT as an Amplifier and switch; Introduction to MOSFET;

Module No. 4	Operation Amplifier (Op-amps)	8 Hours
---------------------	--------------------------------------	----------------

Ideal Op-amp, Differential amplifier: differential and common mode operation, common-mode rejection ratio (CMRR), Practical op-amp circuits: inverting amplifier, non-inverting amplifier, weighted, summer, integrator, differentiator, Large signal operation of op-amps;

Module No. 5	Electrical Machines	7 Hours
---------------------	----------------------------	----------------

DC Machines: Analogy between electrical & magnetic circuits, Types of electrical machines, construction, working principle, and laws. EMF equation of generator, Torque equation of motor.

AC Machines: Single phase transformers – working principle, EMF equation, Voltage transformation ratio, types, and construction; Different loss in transformer

Text Books

1. Charles K Alexander, Mathew N O Sadiku (2012), Fundamentals of Electric Circuits, 5th edition, Tata McGraw Hill.
2. S K Bhattacharya, Basic electrical and electronics engineering, 2nd edition, 2018, Pearson.
3. William H. Hayyt, E. Kemmerly and Steven M. Durbin “Engineering circuit analysis” 8th edition, McGraw Hill 2013
4. Gayakwad Ramkant A., “Op-Amps and Linear Integrated Circuits” 8th edition, PHI Learning, new Delhi, 2011

References

1. S. B. LalSeksena, Kaustuv Dasgupta, Fundamentals of Electrical Engineering, Cambridge University Press, 2016.
2. John Bird (2010), Electrical circuit theory and technology, 4th edition, Newnes publications.
3. Fitzgerald, Higgabogan, Gabel (2009), Basic Electrical Engineering, 5th edition, McGraw Hill.
4. Franco Sergio, Design with Operational Amlifier and Analog Integrated Circuit, McGraw Hill, New Delhi, 2014

List of Experiments:**Module-1:**

1. Colour code of resistors and verification of series and parallel combinations of R, C.
2. Verification of Ohms Law.
3. Verification of KCL and KVL.
4. Verification of voltage and current division principle.
5. Verification of superposition theorem for DC circuits.
6. Verification of Thevenin's theorem for DC circuits.
7. Verification of maximum power transfer theorem for DC circuits.

Module-2:

8. Measurement of power and calculation of power factors with the two-wattmeter method.
9. Verification of voltage, current, power relations for star-delta connected loads.

Module-3:

10. Calculation of two-port network parameters.

Module-4 & 5:

11. Study of characteristics of semiconductor diodes, transistors.
12. Design and simulation of various rectifier circuits.
13. Zener diode as a voltage regulator
14. Design and implementation of the MOSFET voltage amplifier.

Mode of Evaluation	CAT 1 (20%), CAT 2 (20%), CAT 3 (20%), Lab Assessment (40%)
Recommended by the Board of Studies on	